

Exam 1 - AGST 50104

AUTHOR

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Directions:

- A single **PDF** with all your text, code, and figures should be submitted on Blackboard by **03:15 PM** for it to be graded.
- For this exam we will utilize $\alpha = 0.05$.
- Use 2 decimals in any calculations.
- In interpretations, write a maximum of 7 lines (in the resulting PDF), **I will not read any text after the 7th line.**

1) The mean wood volume per tree ($\text{m}^3 \times 10^4$), obtained in an tree breeding program is presented in the "forestry_data2.csv". This experiment evaluated 10 cultivars evaluated in three replications each. For each of the three replications a different randomization was conducted such that all the 10 cultivars were present in each rep. Using the dataset provided, identify if there are any differences in terms of wood volume per tree and if the answer is yes, can you identify the best cultivar?

```
tree <- read_csv("forestry.csv")
```

2) What are the steps required for running a RCBD experiments, i.e., list what would be your pipeline for running an experiment to identify the best irrigation (systems A, B, and C) and fertilizer (N, P, K) regarding grain yield in rice.

3) Interpret the following results:

a. Shapiro-Wilk's test

Shapiro-Wilk normality test

data: residuals

W = 0.99265, p-value = 7.25e-05

b. Levene's test

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	9	1.8503	0.1207
	20		

c. Tukey 1 degree of freedom (additivity test)

```
Warning in aov(height ~ Fertilizer + Cultivar + Error(Fertilizer:Cultivar), :  
Error() model is singular
```

```
dae::tukey.1df(dfaov, error.term = "Fertilizer:Cultivar", data = df)
```

```
Warning in aov(attr(aov.obj, "terms"), data.temp): Error() model is singular
```

```
$Tukey.SS
```

```
[1] 784
```

```
$Tukey.F
```

```
[1] 29.03704
```

```
$Tukey.p
```

```
[1] 0.116813
```

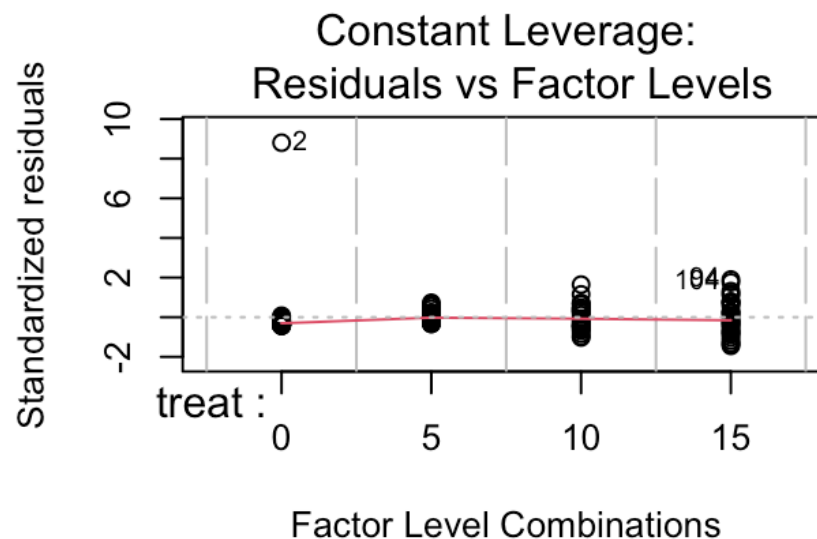
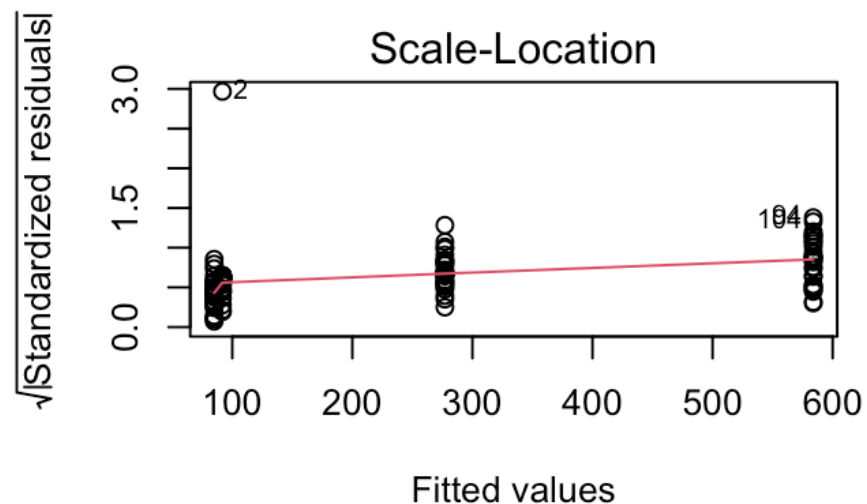
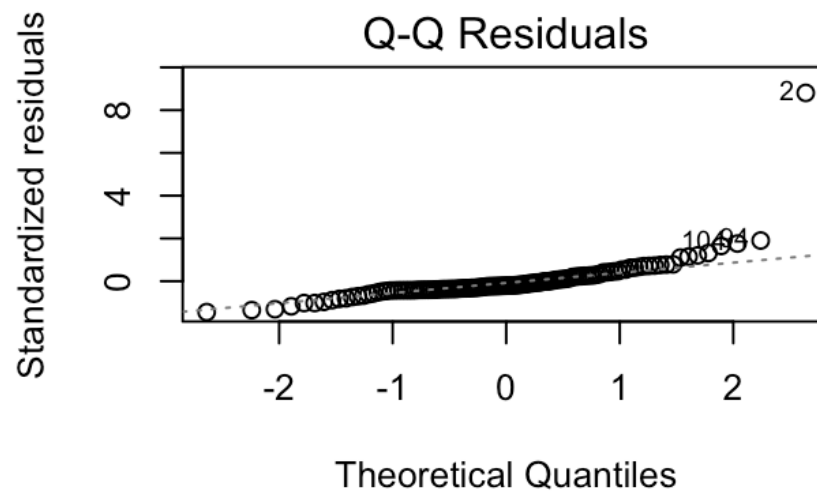
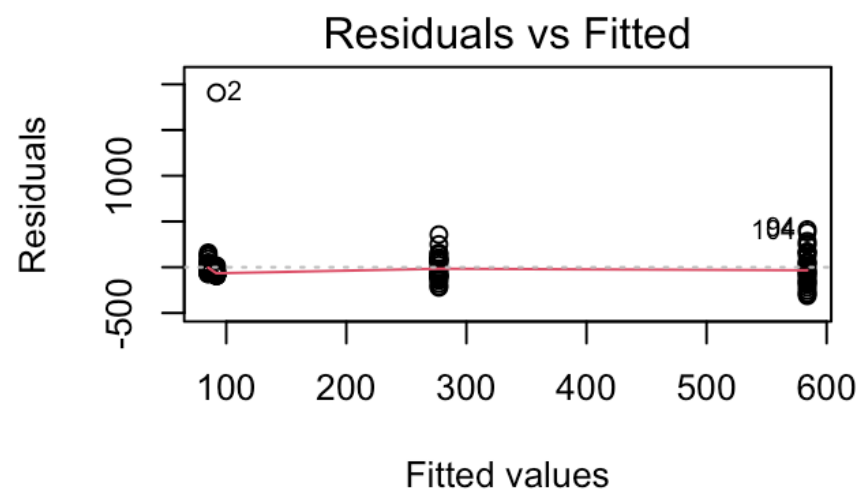
```
$Devn.SS
```

```
[1] 27
```

4) Answer the following questions:

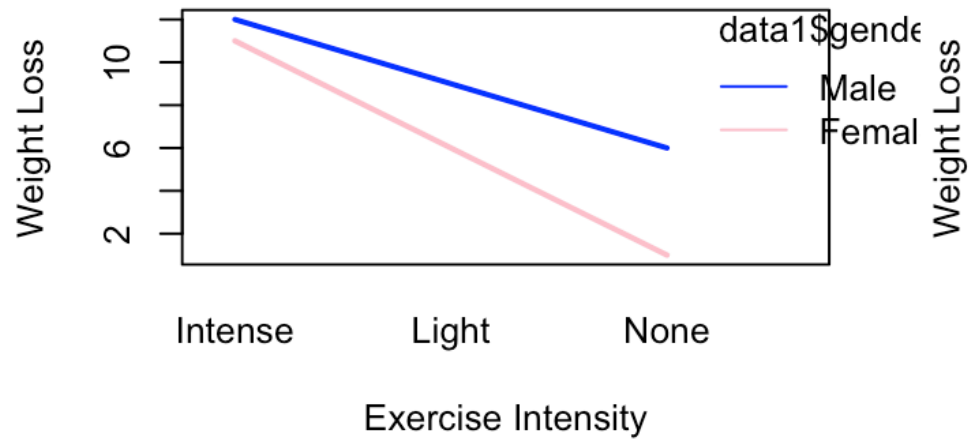
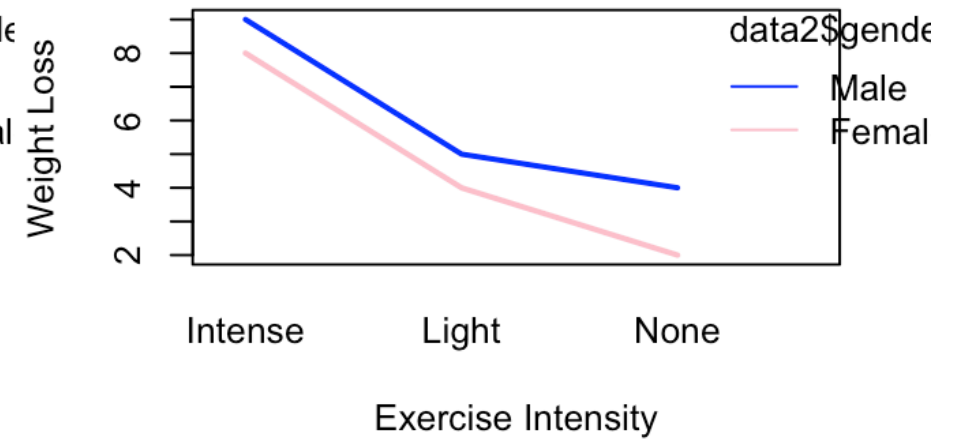
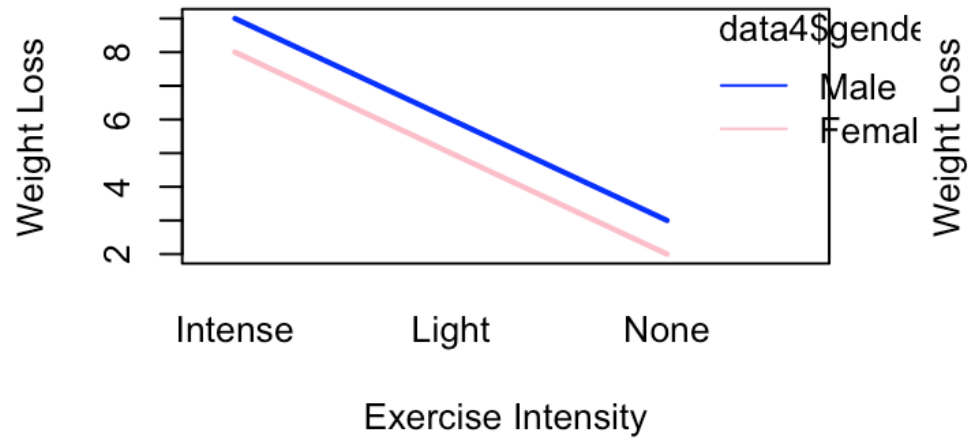
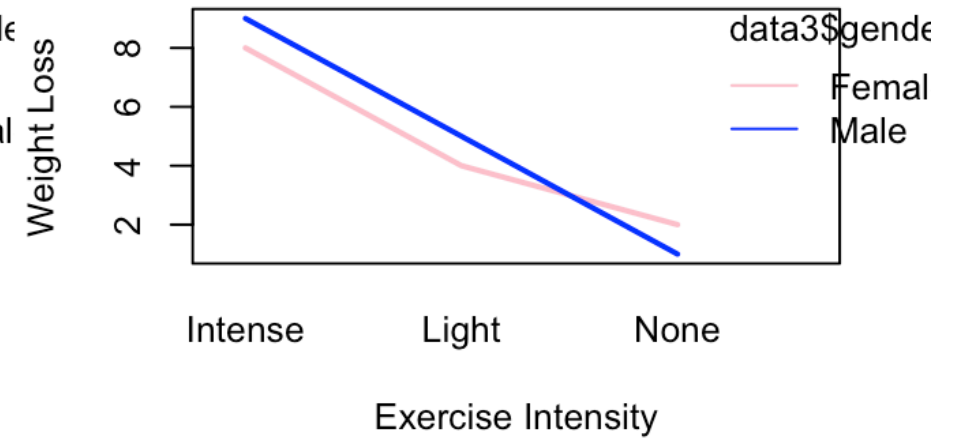
- a. What are the principles of experimentation?
- b. What are the ANOVA requirements/assumptions?

c. Based on the diagnostic plots below, are we safe using ANOVA? If not, what is the problem and what could be a solution?



5) I suspect protein has an effect in bread rising, thus I'm considering running an experiment to test this hypothesis. At the same time, a colleague mention fiber being a possible factor as well. What is the best strategy to confirm the importance of these variables in bread rising and why?

6) Can we say that there is a significant interaction solely based on the interaction plot? Which of the plot(s) below indicates a possible interaction?

A**B****C****D**

7) Answer the following questions:

- a. If the $F_{\text{calculated}} = 1$ for treatment. What would you expect for your pvalue, will it be significant or not?
- b. Testing cultivar effect (30 cultivars) in an anova. What are the null and alternative hypothesis?
- c. In an anova where block is not significant, but I included block in your model. Should you keep block or remove it from your analysis?

8) Interpret the following results. What is the best treatment (assume a greater value of Y is better):

```
q5_aov <- aov(Y ~ treat, data = ex1q5)
summary(q5_aov)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
treat	5	1052.7	210.54	12.06	3.13e-05 ***
Residuals	18	314.2	17.46		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
tukey(ex1q5$Y, ex1q5$treat, 18, 132.2, alpha = 0.05)
```

Tukey's test

Groups	Treatments	Means
--------	------------	-------

a	1.99	84.75
---	------	-------

ab	0.37	82.75
----	------	-------

bc	0.51	77
----	------	----

c	0.71	75
---	------	----

c	1.02	71.75
---	------	-------

d	1.4	65
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9) Below there is the analysis of an experiment that measured the effect of different herbicides on different crops. This was a CRD. Based on the ANOVA table for the sums of squares type II, answer the following questions:

ANOVA Table (Type II tests)

Source	DF	SS	MS	F	P-value
crop	1	455		45.5	
herbicide	3	1333		44.43	
Error	66	660			

a. Fill in the table for the sums of squares type I.

ANOVA Table (Type I tests)

Source	DF	SS	MS	F	P-value
crop	1	436	436	43.6	0.00006
herbicide	3				
Error	66				

b. Interpret the ANOVA results.

10) Design an experiment to evaluate the effect of fertilizer (0kg, 100kg, and 200kg), irrigation (2mm and 5mm), and cultivar ("A", "B", "C") on rice yield. Assume there is space for 72 experimental units and there is a gradient such that the first experimental unit is more fertile than the last. Include the resulting table with the design below.